

Requirements / Specification Document

TITLE: Small Uncrewed Aircraft System

Requesting Laboratory / Division / Group:

Communications Technology Laboratory
Public Safety Research Division (671.00)

General Statement of Need:

The Public Safety Research Division is in need of a lidar mapping, indoor-flight capable small UAS (sUAS) to develop indoor flight configurations and standard test methods for link loss research in our Public Safety Innovation Technology Center. This aircraft will be used for laboratory demonstrations and scientific research of first responder UAS. PSCR desires to purchase a specific sUAS that meets the required specifications.

Item Specifications and Quantities:

Item 1:	Small Uncrewed Aircraft System Airframe and Operator Control Unit	
Quantity:	1 EACH	
Specifications:	1.1	Multicopter sUAS capable of indoor flight in confined spaces, no more than 0.6 meter (2 feet) wide
	1.2	Durable construction, capable of surviving crashes with minimal repair, such as propeller replacement
	1.3	Propeller guards
	1.4	Operational in GPS-denied position-hold
	1.5	Autonomy including real-time object awareness and floor-plan creation
	1.6	Onboard white and blue LED Lights
	1.7	Crash recovery mode (turtle mode)
	1.8	Lidar mapper and forward-facing tracking cameras
	1.9	Ruggedized hard storage case, custom fit for the sUAS, controller and additional batteries
	1.10	Additional spare propellers or accessory kit
	1.11	Two batteries, power supply, and USB cables
	1.12	NDAA Compliant (American Security Drone Act of 2023)
	1.13	Flight time between 15 to 20 minutes

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	1.14	4K Video Recording Resolution
	1.15	Gimbal stabilization with light modes
	1.16	4K minimum Video resolution
	1.17	FLIR camera
	1.18	2-way Audio communications
	1.19	Glass Breaker
	1.20	Payload Dropper
	1.21	Robot Mount
Item 2:	Warranty/Training	
Quantity:	1 EACH	
Specifications:	2.1	3-year support and warranty plan
	2.2	Accessory Kit with spare parts and tool kit, Unlimited Repairs and Maintenance, 1x Airframe Replacement Per Year, On-demand training
	2.3	In-person training
	2.4	Visualization/Operation software with storage for 3 years

Vendor Requirements

1. Possess at time of award/contracting an Export Control Classification Number (ECCN).
2. Possess at time of award/contracting a Unique Entity Identification Number (UEI) which registration is listed as "Active" on the GSA System for Award Management (SAM.gov).

Procurement Support Information and Ancillary Services:

[Provide input for each section Below or Mark Not Applicable (N/A)]

Section	Requirements/Specifications
Planning	☐ List any visits required to allow potential vendors to see site specific constraints.
Considerations	☐ US citizen required on-site (foreign nationals require pre-registration 30 days prior to visit.)
☒ N/A	[Explain Requirements.]

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- ☐ Vendor Representative On-Site more than three (3) days.
- ☐ Other Considerations:
[List other considerations to ensure an accurate quote.]

Shipping &
Delivery
☐ N/A

Shipping Address:

- ☒ NIST-Boulder
ATTN: Donald Harriss
Mail Stop: 671
325 Broadway
Boulder, CO 80305

- ☐ NIST-Gaithersburg
ATTN:
Mail Stop:
100 Bureau Dr
Gaithersburg MD 20899
- ☐ [Alternate Shipping Address]

Shipping Criteria:

- ☒ Partial Delivery Acceptable
- ☐ [Enter Shipping and Delivery information by item.]
- ☐ Direct Delivery to OU Building Required
[Explain Need for Requirement]
Building /Room Number: [Enter Building/Room Number]
☐ Building has Loading Dock
Containerization Preference: [Enter Container Type]
Other Requirements:
[List Any Other Requirement]

Delivery Date Criteria:

- ☒ Delivery Shall Be Completed No Later Than:
7/19/2026
- ☐ Delivery Shall Be Made No Earlier Than:
Click or tap to enter a date.
- ☐ Scheduled Deliveries:

Item Number	Due Date

- ☐ Other:

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[List Other Shipping, Delivery or Special Requirements.]

Electronic Media

[Enter Electronic Media Requirements by Item.]

☒ N/A

☒ **Software**

☐ Electronic Manuals

☐ License Key

☐ Other:

[List Any Other Electronic Media Requirements]

☐ US Government Email Address for Software Delivery:

[Enter Address]

Installation

[Enter Installation Requirements by Item]

☒ N/A

☐ Vendor Technician on Site

☐ Rigging

☐ Uncrating / Unpackaging

☐ Removal of Packaging Material

☐ Equipment Set Up

☐ Start-Up Services

☐ Turnkey Installation

☐ Other:

[Enter Details]

NOTE: TPOC shall be the coordination POC.

Facility / Utility Considerations

[List Site Specific Utility Considerations by Item]

☒ N/A

☐ OFPM Work Order Numbers

[List Work Order Numbers]

☐ Power Available

Voltage: [Enter voltage] (V)

Maximum Current: [Enter max current draw] (A)

Phase: [Choose Phase]

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Frequency: [Typically 60Hz] (Hz)
NEMA Plug Type: [Enter Plug Type] (i.e.: NEMA L5-20P)
Environment: [Choose Environment]
Other Power Considerations:
[List Any Other Power Considerations.]
☐ Utility Connections:
[List Relevant Available Utilities, Pressure(s) and Flowrate(s).]
☐ Other Considerations:
[List Other Considerations to Ensure an Accurate Quote.]

Inspection and
Acceptance
☐ N/A

[Inspection Requirements Per Item (Who, When, What, Where)]

- ☐ On-Site
☒ Acceptance Not Expected to Exceed 7 Business Days
☐ Acceptance Expected to Exceed 7 Business Days:
[Duration]
☐ Inspection and Acceptance Plan:
[Explanation]
☐ Other:
[Enter Other Requirements]

Training
☐ N/A

Training Requirements (Who, When, What, Where)

- ☒ Number of Trainees: Up to three
☒ On-Site
☐ Off-Site Location: [Enter Location(s)]
Is Trainee Travel Required: NO

Scope of Training

- ☒ Operation ☒ Maintenance ☒ Troubleshooting
☒ Safety ☐ Other

[Enter Other Scope of Training Requirements]

- ☐ Training Materials to be Provided by Government:
[Describe Materials]
☐ Training Materials to be Provided by Vendor:

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[Describe Materials]

☐ Other

[Enter Other Training Requirements]

NOTE: SME shall be coordination POC.

Warranty

☐ N/A

Base Manufacturer Warranty is acceptable

Warranty Length: 3 YEARS

What is included?

☒ Labor ☒ Parts ☐ Travel

☐ Cost and Liability for Returns

☐ Other

[Enter Other Requirements]

☐ 1 Option Period

☐ 2 Option Periods

☐ 3 Option Periods

☐ 4 Option Periods (Maximum)

Calibration

☒ N/A

[Describe Calibration Requirements Per Item]

Calibration Plan Length: [i.e. 3 - Years]

What is Included?

☐ Labor ☐ Spare Parts ☐ Consumables

☐ Travel ☐ Cost and Liability for Shipment

☐ Software Updates ☐ Manuals

☐ Other

[Enter Other Requirements]

☐ Requirements:

[Enter Requirements] (i.e.: On-Site Within 72 hours, etc.)

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- ☐ 1 Option Period
Length: [i.e. 1 – Year / 6 - Months]
- ☐ 2 Option Periods
- ☐ 3 Option Periods
- ☐ 4 Option Periods (Maximum)

Maintenance

☐ N/A

[Describe Maintenance Requirements Per Item]

Maintenance Plan Length: 3 YEARS

What is Included?

- ☒ Labor ☒ Spare Parts ☐ Consumables
- ☐ Travel ☐ Cost and Liability for Shipment
- ☒ Software Updates ☒ Manuals
- ☒ Other
1x Airframe replacement per year
- ☒ Requirements:
Unlimited repairs and maintenance

- ☐ 1 Option Period
Length: [i.e. 1 – Year / 6 - Months]
- ☐ 2 Option Periods
- ☐ 3 Option Periods
- ☐ 4 Option Periods (Maximum)

Government Furnished Material / Equipment

☒ N/A

List Any Government Furnished Material / Equipment Per Item]

- ☐ Samples for Testing
- ☐ Equipment which will leave Government Site
Description: [Enter Description]
Serial Number: [Enter Serial Number]
NIST Property Number: [Enter NIST Property Number]
- ☐ Other: [Enter Other Details]

Travel

☒ N/A

[Describe any Travel Requirements Per Item]

- ☐ Explain:
[Enter Justification / Requirement]
- ☐ Other:
[Enter Other Requirements]

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NOTE TO VENDOR: To the maximum extent possible, include travel requirement estimates as part of pertinent sections above (Installation, Maintenance, Repair, etc.). Otherwise, Travel will be a separate line item, reimbursable per the Federal Travel Regulation.